Online Survey System

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System Design/Info-C450

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SYSTEM SPECIFICATION DOCUMENT
Online Survey System

1. CUSTOMER PROBLEM STATEMENT

A customer with student organizations or event management as their main focus will need quick feedback from participants, members, or attendees. The current way of getting feedback is slow and vastly inefficient. Different tools are used to create surveys, links are distributed through various channels, and reminders are sent manually.

The customer wants to quickly create surveys with common question types, publish them, and share a single link that works seamlessly on both mobile phones and computers. The system will also support anonymous feedback and be very open about the data collected. After the responses are submitted, the system will securely store the data, send a notification to the respondent that the submission was received, and also provide summary results that are easy to understand. The customer wants a download option so that the results can be downloaded and reported without any extra manual work.

2. GLOSSARY OF TERMS

Term	Definition
Survey Creator	Person who creates and publishes a survey.
Respondent	Person who opens the link and submits answers.
Anonymous Mode	Setting that avoids storing identifying information.
Publish	Making the survey available to respondents.
Share Link	URL that opens the survey.
Response	A completed set of answers submitted by a respondent.
Close Date/Time	Deadline after which the survey no longer accepts submissions.
Export	Download of survey results in CSV format.

3. SYSTEM REQUIREMENTS

Functional Requirements

No.	Priority Weight	Description
REQ-1	High	The system will allow a creator to create a survey with a title and description.
REQ-2	High	The system will support question types: multiple choice, short answer, and rating scale.

REQ-3	Medium	The system will allow the creator to mark questions as required.
REQ-4	Low	The system will allow the creator to set a close date/time for the survey.
REQ-5	High	The system will publish the survey and generate a share link.
REQ-6	High	The system will allow respondents to open the share link and submit one response.
REQ-7	Medium	The system will provide an anonymous mode option for the survey.
REQ-8	High	The system will display basic results summaries for each question.
REQ-9	High	The system will export results to CSV for customer.

Nonfunctional Requirements (FURPS)

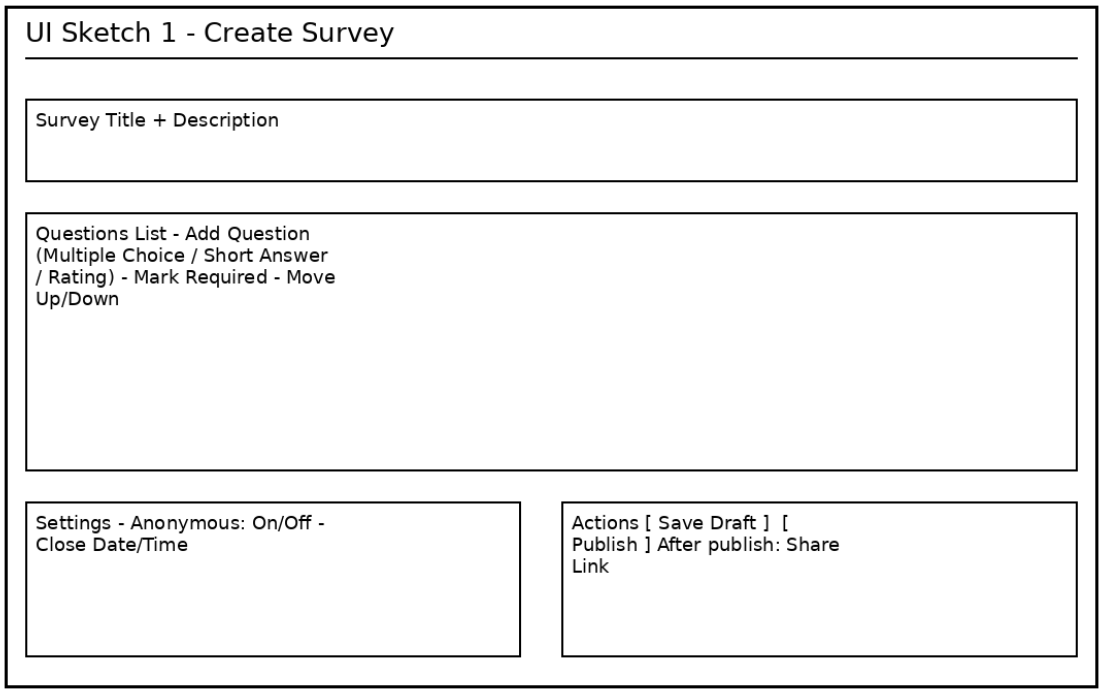
No.	Priority Weight	FURPS Category	Description
NFR-1	High	Usability	Respondents must be able to complete a short survey on a phone without zooming or horizontal scrolling.
NFR-2	High	Reliability	Each submitted response must be saved and confirmed to the respondent.
NFR-3	Medium	Performance	Results summaries should load quickly for typical class or club survey sizes.
NFR-4	High	Functionality	Anonymous mode must clearly communicate what is and is not collected.
NFR-5	Medium	Supportability	CSV exports must open correctly in common spreadsheet tools.

4. USER INTERFACE REQUIREMENTS

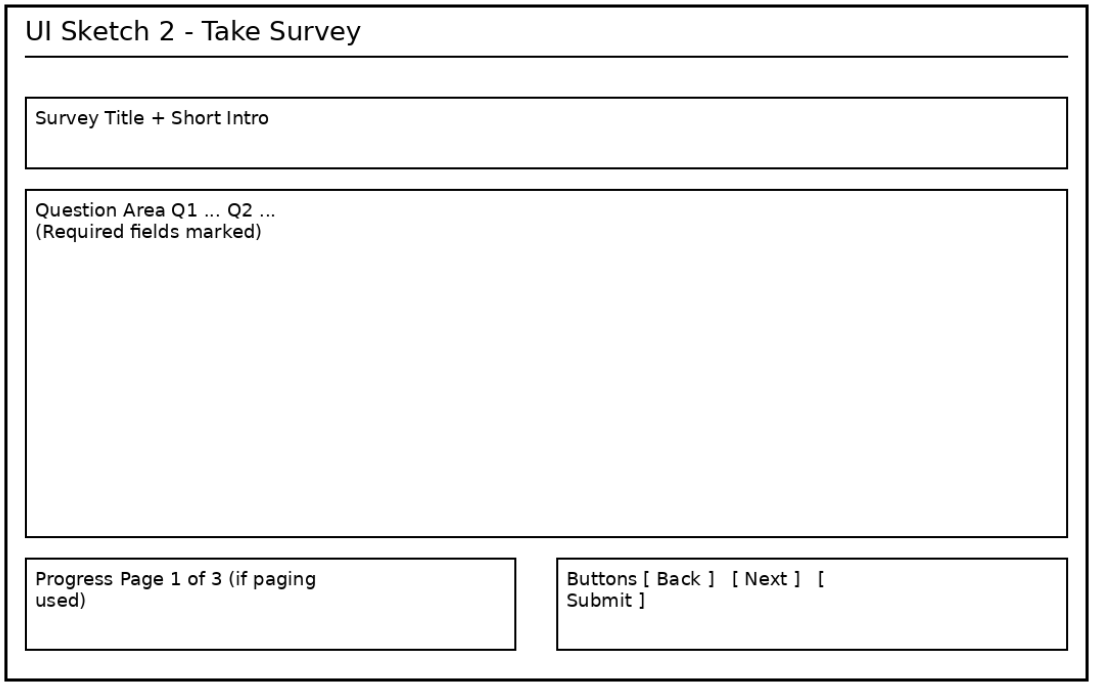
No.	Priority Weight	Description
UI-1	High	Create Survey screen allows adding questions and setting anonymous mode.
UI-2	High	Take Survey screen shows questions clearly and allows submit with confirmation.
UI-3	High	The results screen shows summaries and provides a CSV export for the customer only.

Sketches (conceptual wireframes):

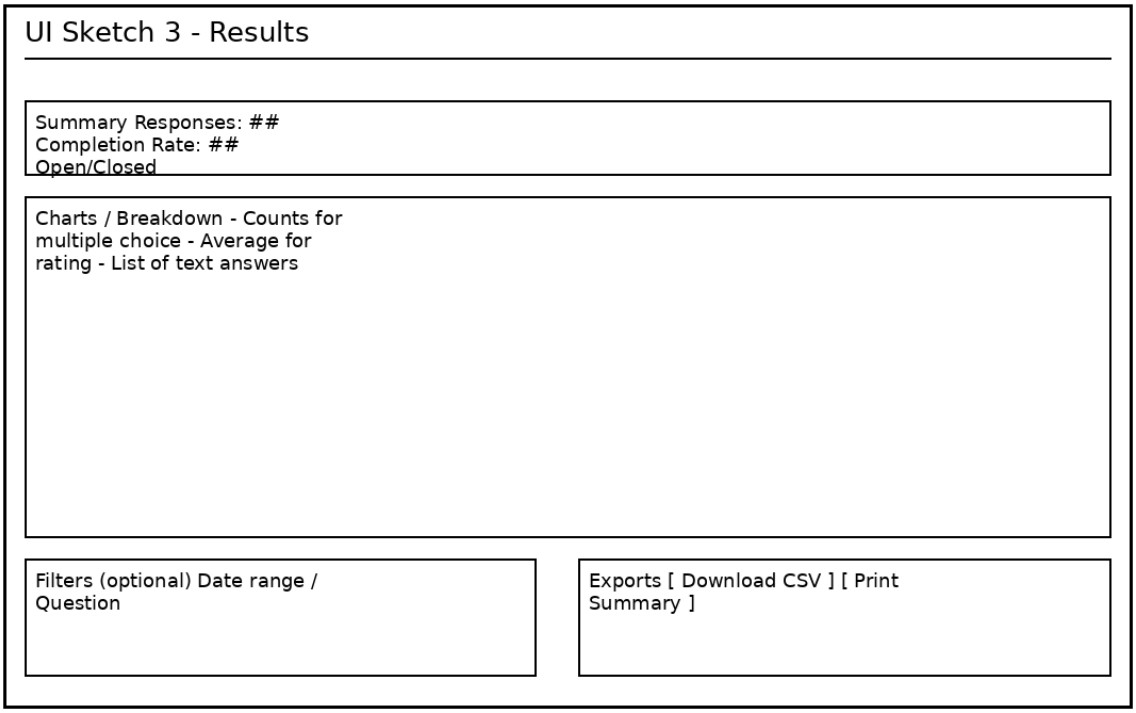
UI-1: Create Survey screen allows adding questions and setting anonymous mode and close date/time.



UI-2: Take Survey screen shows questions clearly and allows submit with confirmation.



UI-3: Results screen shows summaries and provides CSV export.



5. PLAN OF WORK

Status values: Done, In Progress, Not Started

Step	Work Item	Status	Notes
1	Confirm scope and target users	Done	Kept the project focused on survey creation, link sharing, and results export.
2	Set up development environment	Done	Environment and repository prepared.
3	Define requirements and UI sketches	Done	Captured core requirements and key screens.
4	Design data model (Survey, Question, Response)	Will start next week	Define entities and relationships.
5	Implement create/publish/share link flow	Not Started	Core creator workflow.
6	Implement respondent submit + close date enforcement	Not Started	Submission and deadline logic.
7	Implement results summary + CSV export	Not Started	Reporting and export.

8	Testing and final documentation	Not Started	Test and finalize deliverables.

Functional Requirement Specification

1. Stakeholders

- Customer (Survey Owner): creates surveys, publishes links, views anonymous results, and exports CSV/Excel.
- Respondents (Participants): fill out surveys quickly on mobile/desktop without creating accounts.
- Instructor/Sponsor: evaluates whether the system meets the assignment requirements and novelty goals.
- IT/Hosting Provider (AWS): hosts the system, provides reliability, backups, monitoring.
- Email Provider (optional): sends invitations/reminders if the customer chooses email distribution.

2. Actors and Goals

Primary actors

- Customer (Survey Owner): Create and manage surveys, publish/close them, share links, view anonymous results, and export results to CSV/Excel.
- Respondent: Open the survey link, complete the survey, and submit responses anonymously.

Secondary actors

- Email/Notification Service (optional): Send invitations/reminders when the customer chooses email distribution.
- System: Stores anonymous responses, validates required questions, aggregates results, and generates exports.

3. Use Cases

Customer (Survey Owner)

- Create survey: Create a new survey with a title/description and save it.
- Edit survey: Update survey details and questions before publishing.
- Add questions: Add question types (multiple choice, rating scale, short answer).
- Set required questions: Mark questions as required and enforce validation rules.
- Publish survey & generate link: Publish and generate a shareable public survey link.
- Share link: Copy/share the public link.
- Close survey: Stop accepting responses while keeping results available.

- View anonymous results dashboard: View response counts and summary charts without identifying respondents.
- Export results: Export responses to CSV/Excel.

Respondent

- Open survey link: Access the survey from a public link on any device.
- Complete survey: Answer questions using a clear, mobile friendly form.
- Required field validation: Prevent missing required answers before submission.
- Submit anonymous response: Submit once and receive a confirmation message.

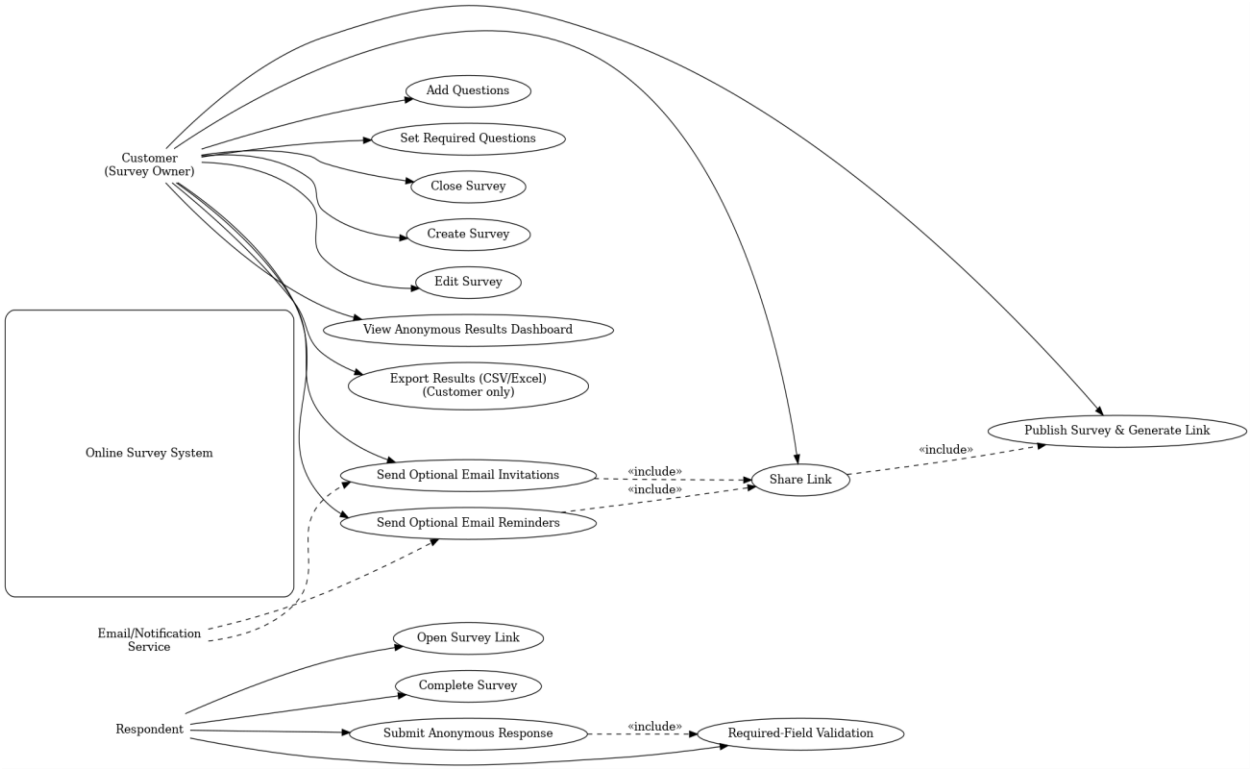
Email/Notification Service (this can be optional)

- Send email invitations: Customer provides emails, and the system sends a message with the survey link.
- Send email reminders: Customer triggers reminders to the same list.

System

- Anonymous response storage: Store responses without collecting identity.
- Results aggregation: Compute summaries.

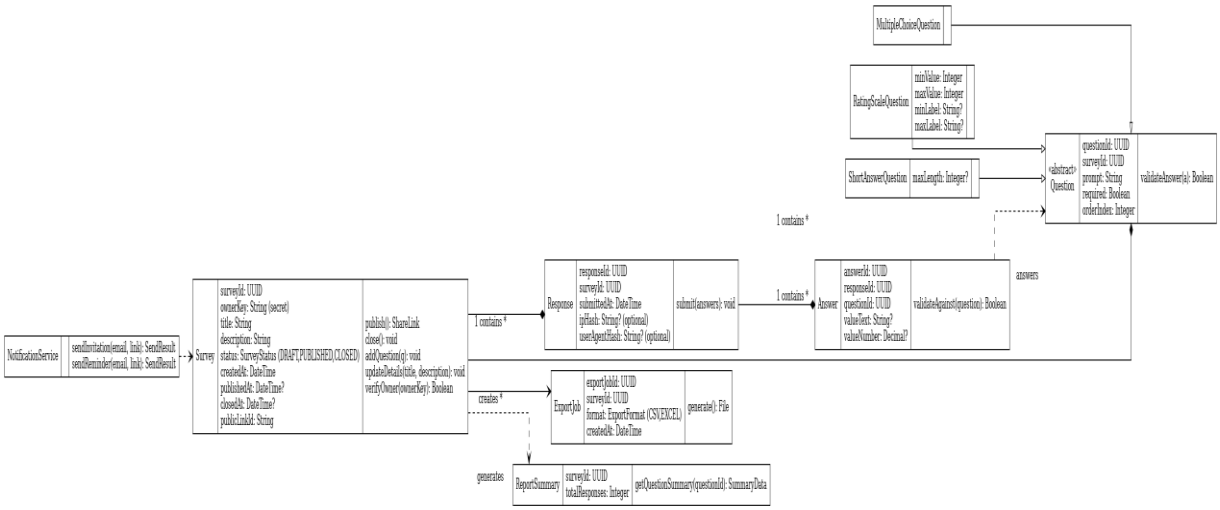
4. Use Case Diagram



Relationship comments

- Submit anonymous response «include» Required field validation
- Share link «include» Publish survey & generate link
- Send optional invitations «include» Share link
- Send optional reminders «include» Share link

5. Class Diagram



Problem statement

Small organizations, campus groups, faculty members, and local businesses, for example, often use surveys to collect feedback. But generally, the whole operation is divided among spreadsheets, shared links, and manual follow ups. It results in low response rates, duplicate or incomplete responses, data cleaning issues, and delayed reporting. The goal is a single, straightforward system that helps users plan a questionnaire, reach the right audience, collect feedback securely, and have results quickly.

A brief description

The Online Survey System enables survey creators to develop surveys using the most popular question types. They can publish the survey as a link or send invitations to a list of participants and then gather responses in a database. Creators will be able to view summaries and simple charts, and export results to Excel. Survey takers will have the flexibility to complete surveys on a phone or a computer.

Objectives of the system

The objectives are to make surveys faster to create and manage, improve response rates through invitations and reminders, reduce duplicate and incomplete submissions, and provide quicker reporting for decision making.

System requirements

The system supports Admin, Survey Creator, and Survey Takers roles.

Users can do the following:

- Create an account, log in, and reset a password
- Create, edit, publish, and close surveys
- Add multiple choice, rating scale, and short answer questions
- Save surveys as drafts and publish with a shareable link
- Send invitations and reminders to non responders
- Require answers for selected questions
- Limit invite only surveys to one submission per invited respondent
- View response summaries and basic charts
- Export results to Excel
- Complete surveys on mobile or desktop

Typical customers

- Faculty or teaching assistants collecting course feedback
- Student organizations running event surveys or polls
- Small businesses collecting customer satisfaction feedback
- Nonprofits collecting volunteer or community input

Project planning

- **Software:** Web browser (Chrome), IDE (Eclipse), server environment, database software, email service for invitations/reminders, and AWS as an option for cloud hosting and deployment.
- **Hardware:** PC/laptop and optional cloud resources if deployed on AWS.
- **Network:** Reliable internet connection for development, testing, and cloud access.
- **Front end:** JavaScript
- **Back end:** Node.js
- **Database:** MySQL

Development plan

- **W1-W2:** Confirm requirements with target users, design database, set up project structure, connect front end and back end database
- **W3-W4:** Build login/registration, user roles, and survey builder (draft and publish)
- **W5-W6:** Build response collection, required questions, and duplicate prevention for invite only surveys
- **W7:** Add audience upload, invitations, and reminders
- **W8:** Add reporting dashboards and Excel export, midterm demo testing
- **W9-W10:** Have usability testing and improve based on feedback

- **W11-W12:** Finalize features, handle edge cases, and security checks
- **W13-W15:** Write test cases, complete documentation, and record the final demo

System Sequence Diagrams and Activity Diagrams

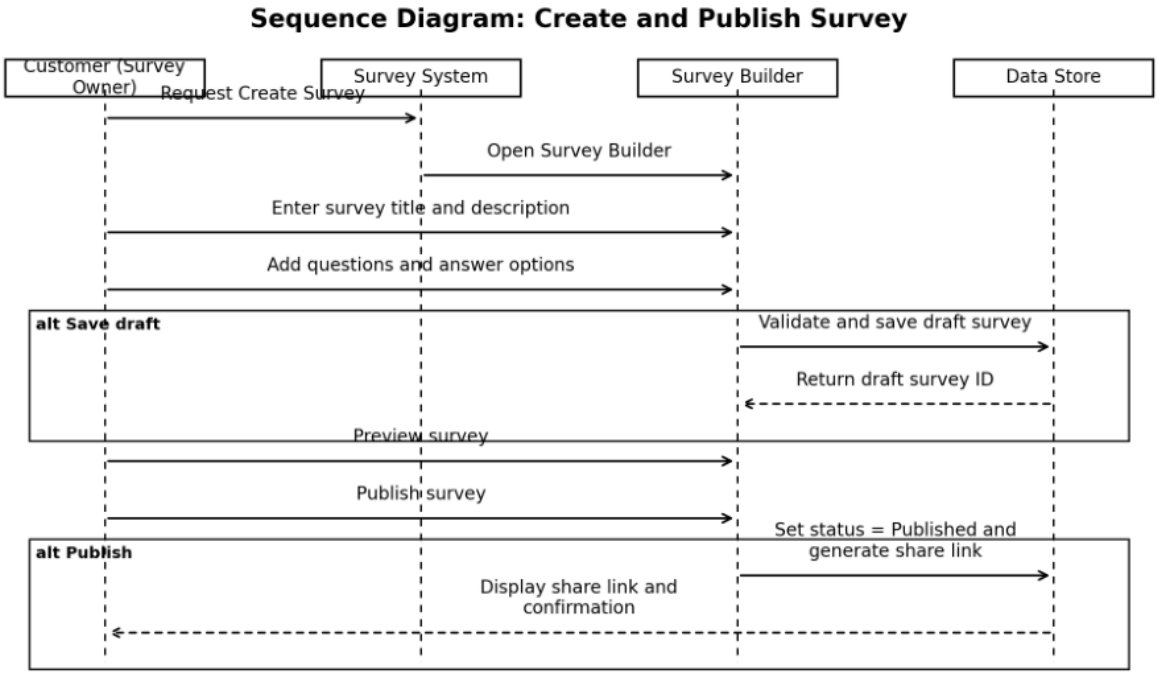
System Sequence Diagrams

Actor and objects in this interaction:

Actor: Customer (survey owner). Objects: Survey System, Survey Builder, Data Store.

Steps of interaction:

1. Customer requests to create a new survey.
2. System opens the survey builder.
3. Customer enters the survey title and description.
4. Customer adds questions and answer options.
5. Survey builder validates the content and saves the survey as a draft in the data store.
6. Customer previews the survey and publishes it.
7. System updates the survey status to Published, generates a share link, and displays confirmation.

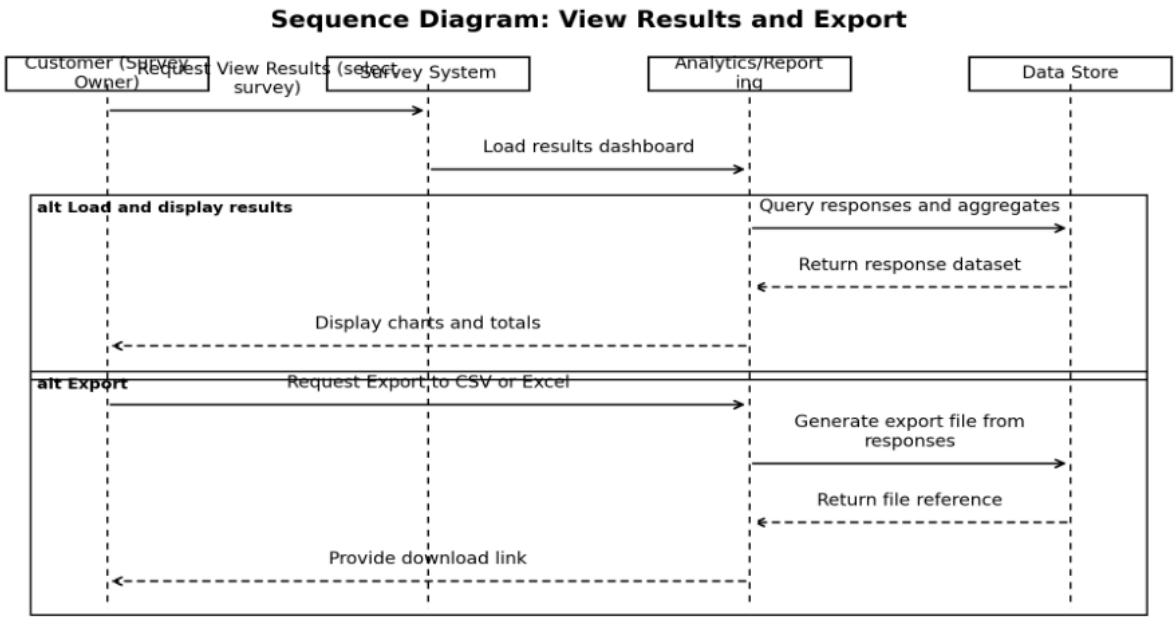


Actor and objects in this interaction:

Actor: Customer (survey owner). Objects: Survey System, Analytics and Reporting, Data Store.

Steps of interaction:

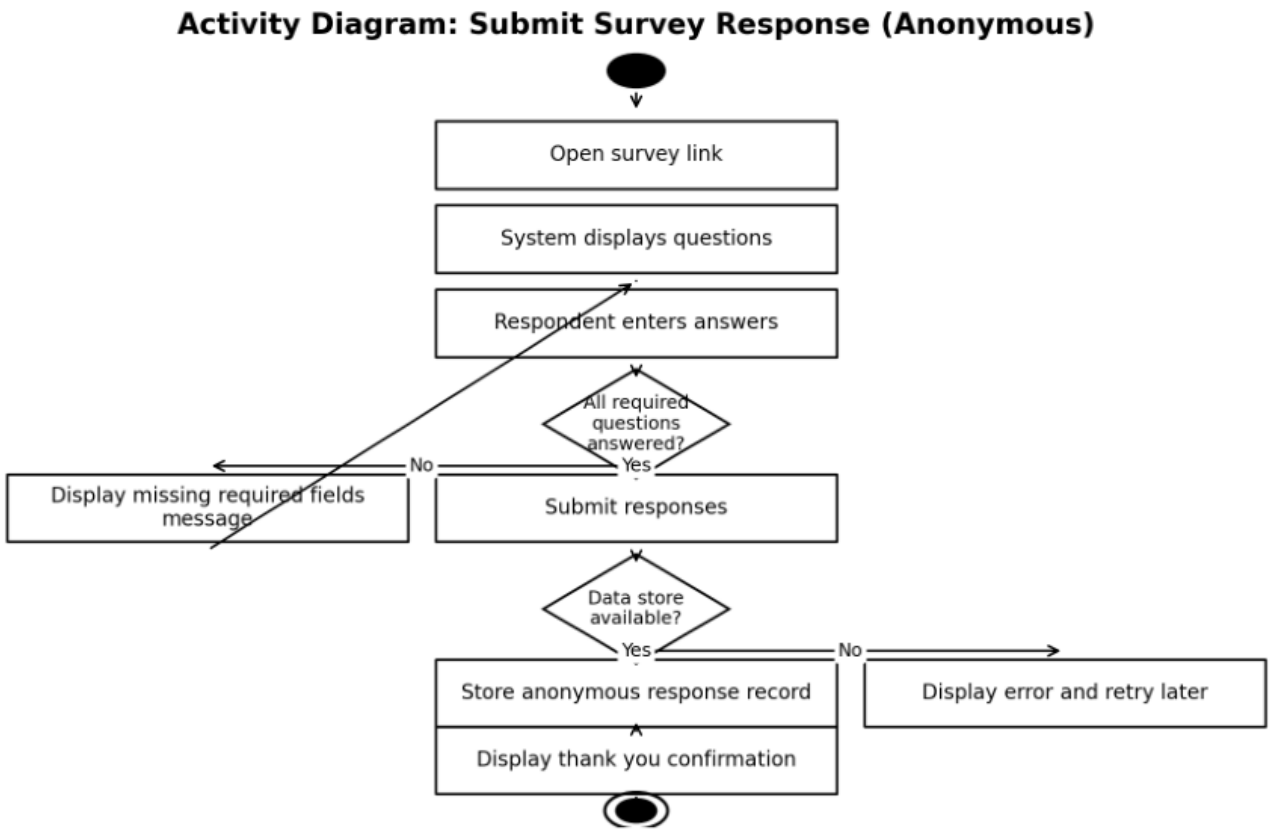
8. Customer selects a survey and requests to view results.
9. System loads the results dashboard and requests response data from the data store.
10. Analytics and Reporting aggregates responses and displays charts and totals to the customer.
11. Customer requests an export in CSV or Excel format.
12. System generates the export file from stored responses and provides a download link.



Activity Diagrams

States: Initial state: A respondent opens the survey link. Final state: The response is stored and a thank you confirmation is displayed, or an error is shown and the respondent is asked to retry later.

Actions: The respondent opens the survey link and the system displays questions. The respondent enters answers. If required questions are missing, the system shows a message and returns the respondent to the survey. If the submission is valid, the system stores the response without collecting identity information and then shows a confirmation.

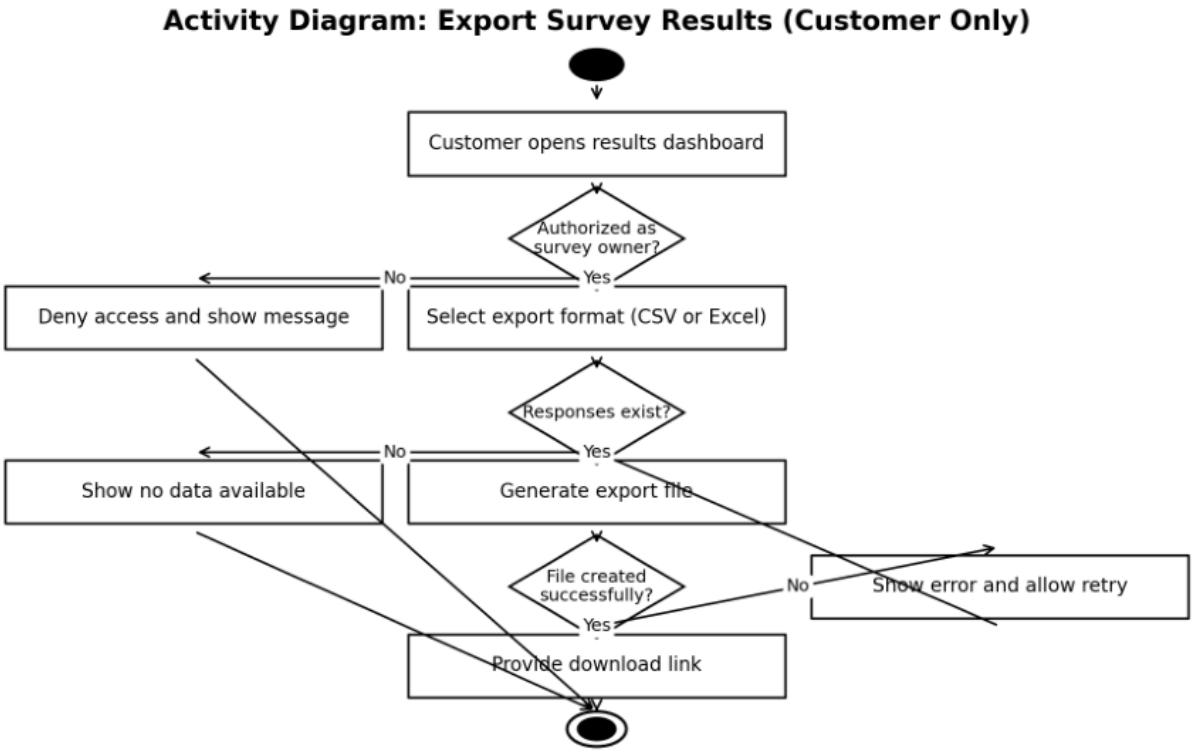


States:

Initial state: The customer opens the results dashboard. Final state: The export file download link is provided, or access is denied, or no data is available.

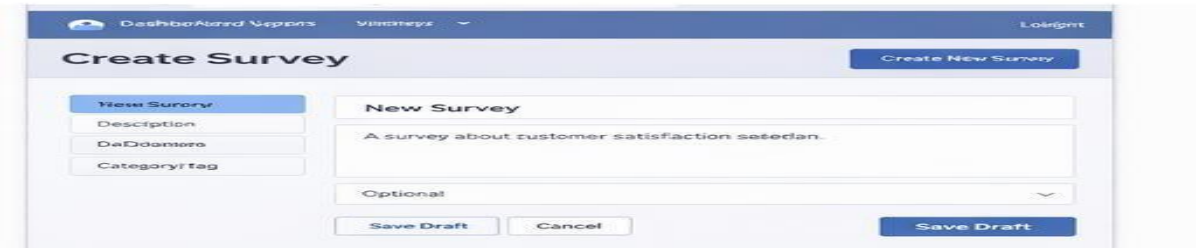
Actions: The customer opens the results dashboard. The system verifies that the customer is the survey owner. If not authorized, the system denies access. If authorized, the customer selects an

export format. If responses exist, the system generates the export file and provides a download link. If file creation fails, the system shows an error and allows retry.



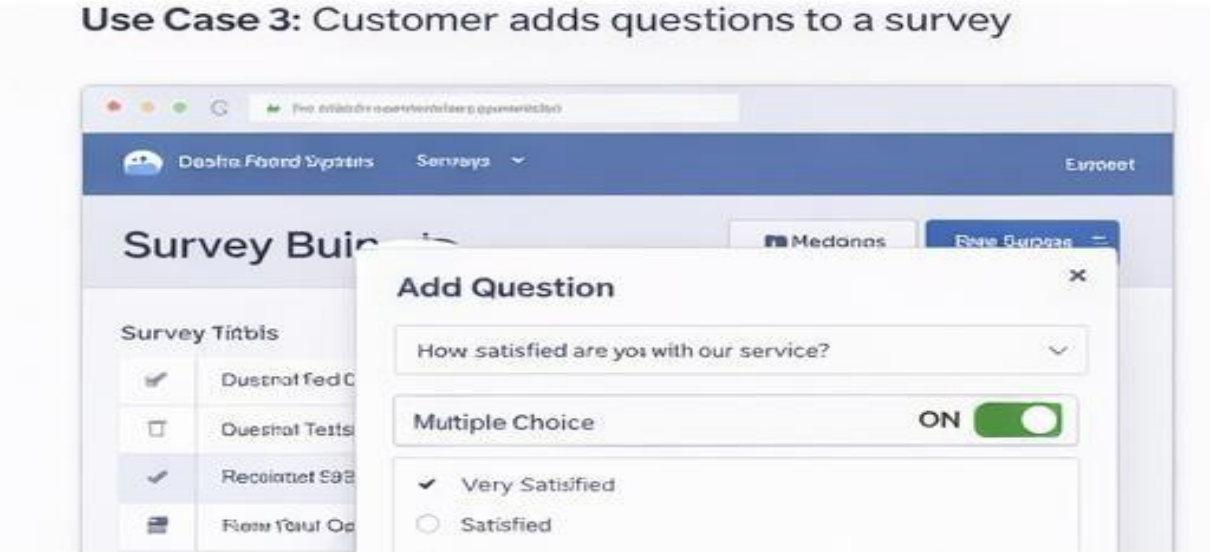
User Interface Specifications

Use Case: Create Survey



Navigation: Surveys → Create New Survey
→ Enter title/description → Save Draft

Use Case: Add Questions



Navigation: Surveys → Edit (Draft) → Add Question → Select Type → Save Question

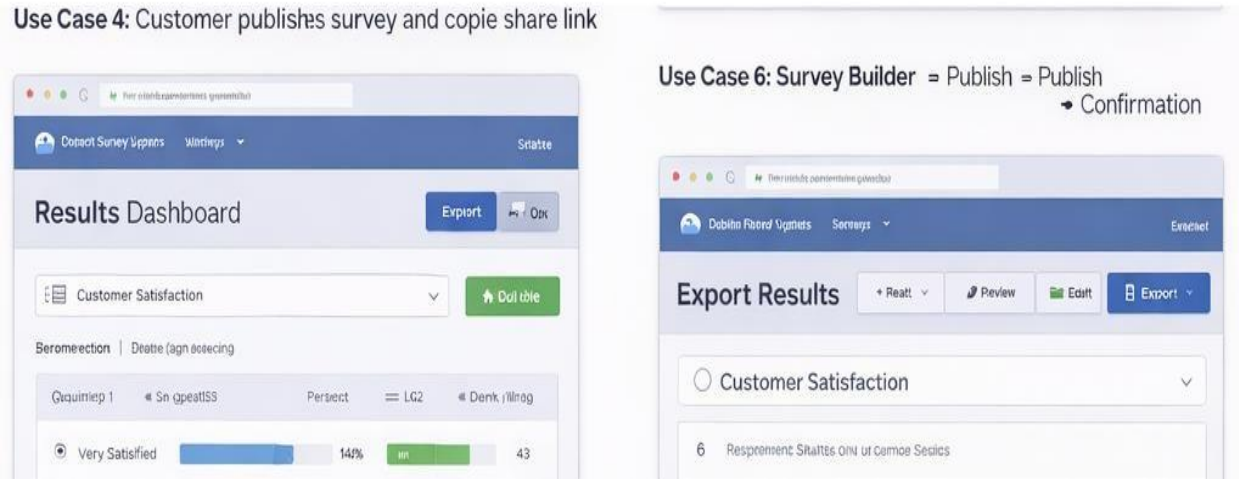
Use Case: Publish Survey & Generate Link

Navigation: Survey Builder → Publish → Confirmation → Copy/Share Link

Use Case: Submit Anonymous Response (Validation)

Navigation: Open public link → Answer questions → Validate required fields → Submit → Confirmation

Use Case: Export Results (CSV/Excel)



Navigation: Results Dashboard → Export → Choose CSV/Excel → Generate → Download

References

Tilley, S., & Rosenblatt, H. (2017). Systems analysis and design (11th ed.). Cengage.